



BUSINESS

PREDICTING HIGH-PROFIT LEADS

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INTRODUCTION

This project uses our lead data to predict which customers are most likely to be very profitable. For each lead we know how they arrived (basic attributes), how much money they brought in (revenue), and how much they cost us (cost). From these, we calculate a margin number: $(\text{revenue} - \text{cost}) / \text{cost}$. A higher margin means better profit relative to what we spent.

OUR VISION

1

To make the task simple, we turn margin into two groups

2

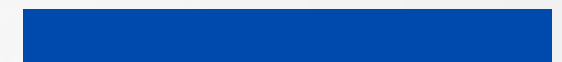
We look at the training margins and find the value that separates the top 25% from the rest

3

Any lead at or above that value is “high profit” (label 1). Any lead below it is “not high profit” (label 0)

4

This gives the model a clear yes/no question: is this lead among the best quarter or not?



MODEL USE & ACCURACY

We then train a Random Forest model. It learns patterns in the lead attributes that are linked to being high profit.

We balance the classes so the model pays attention to both high-profit and not-high-profit cases.

On the test data, the model reaches about 79% accuracy. That means roughly 8 out of 10 leads are labeled correctly as high-profit or not. The confusion matrix shows most decisions are correct, with a small number of missed high-profit leads and some false alarms.

WHAT THIS MEANS FOR BUSINESS IS SIMPLE:

we can quickly flag the leads most likely to deliver strong profit and prioritize them. If we want to include more “good but not top” leads, we can use a softer cutoff or predict the exact margin instead of just yes/no.

